

Modelling long-term electricity load demand for rural electrification planning

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Abstract— Rural electrification plays a key role in terms of achieving the target of universal access to electricity by 2030. In order to deploy reliable and sustainable solutions, the sizing phase of decentralized power systems requires specific inputs such as long-term forecasts of the electricity load. In rural contexts, access to electricity is extremely interlinked to social-economic complex dimensions that in turn, influence the electricity demand. In this paper, we rely on System Dynamics for modelling such complexities, and we calibrate the results with data from a hydroelectric-based electrification programme implemented in 2005 in the rural community of Ikondo, Tanzania. We coupled the model with a newly extended version of a load stochastic tool, namely Lifetime – LoadProGen, in order to introduce a more reliable and calibrated approach to long-term electricity load forecasts.

Index Terms-- rural electrification, electricity demand, system-dynamics, load profiles, electricity-development nexus.

I. INTRODUCTION

Rural electrification is expected to largely contribute to the achievement of electricity access goals in developing countries by 2030, since people still living without electricity will live predominantly in rural areas [1]. When dealing with rural electrification plans, the assessment of long-term electricity demand is one of the most critical and complex steps. Indeed, wrong predictions of electricity demand could negatively impact the local socio-economic development and cause an inappropriate sizing of local energy solutions, leading to supply shortages or cost recovery failure, as argued and demonstrated by Riva et al. [2], Cabral et al. [3], Kivaisi [4], and Hartvigsson et al. [5]. In this context, the socio-economic complexities behind the evolution of the electricity demand in rural areas are hindering appropriate assessments and projections in the long-run. Indeed, “the dynamics of growth and electrification are complex, involving many underlying [socio-economic] forces” ([6], pg. 666). This renders the use of linear or predefined sets of relations of cause and effect to describe the issue inaccurate [7]. Thus,

being able to model aspects and dynamics behind the nexus between electricity demand growth and development can lead to more reliable energy demand projection models and reliable planning strategies for rural electrification.

In this context, the aim of this article is therefore to model the main fundamental dynamics and variables that determine the evolution of electricity demand in rural settings. A system-dynamics-based approach will be used to conceptualise and formulate the main drivers of the evolution of the electricity demand in rural areas, meanwhile a soft-link with a load profile model will allow to integrate both long- and short-term aspects of the electricity demand for energy planning purposes.

II. LITERATURE REVIEW AND METHODOLOGY SELECTION

Despite the existence of many mathematical modelling frameworks, the application of models for long-term electricity demand forecasting in rural contexts is very limited. Riva et al. [8] identified that in rural energy planning literature, most of the studies do not consider any evolution of the energy demand along their planning horizon. Few studies assume a dynamic demand over the years and most of them forecasts its evolution through exogenous and/or arbitrary trends or scenarios, as done in [9]–[11].

In this context, Riva et al. [7] analysed, through causal-loop diagrams, the interrelations between electricity use and socio-economic development in rural areas, confirming that an appropriate modelling framework for generating long-term projections of electricity consumption in rural settings should start from the conceptualisation and formulation of such complex and multifaceted dynamics. Moreover, they suggest that simulations based on Systems Dynamics (SD) can allow quantitative modelling of these complex and dynamic relations. SD has been firstly introduced by Prof. J.W. Forrester, Massachusetts Institute of Technology, in ‘50s, and formalised in his well-known book entitled *Industrial Dynamics* [12]. It is used for understanding, describing, tackling and modelling well-defined endogenous problems

concerning existing systems that are suitable to be formulated as interaction between elements through causal relationships. The modelling framework based on SD resulted to be the most appropriate approach to tackle the aim of this paper, given the multiplicity of cause-effect relations and feedbacks, and time-delays characterizing the nexus between electricity demand and socio-economic development.

III. MATERIAL AND METHODS

A. Model Conceptualisation

This phase consists in describing the behaviour and diagramming the feedback loops of the system. A real case-study is used as reference to build and test the confidence of the model, *i.e.* a hydroelectric-based electrification programme implemented in the rural community of Ikondo, Tanzania, by the Italian NGO CEFA Onlus. The hydropower plant was built in 2005 with a nominal power of 83 kW to supply electricity to the village of Ikondo, through 8 km of MV distribution network. In 2016, CEFA completed the upgrade of the power facility with an additional 350 kW turbine – achieving an overall generation capacity of 433 kW – and the realization of 47 km of MV distribution network that reached other 4 villages in immediate surroundings. Ikondo has a population of approximately 820 households (HHs). The village is characterised by an agriculture-based livelihood, with around 100-120 Income Generating Activities (IGAs) started after the electrification of the community in 2005.

Key variables and causal-loop diagrams for Ikondo village are based on the analysis of the energy-development nexus carried out by Riva et al. in [7], and through three on-field interviews conducted in 2006, 2017, and 2018. The representation of the causal-loop diagrams consists of variables linked by arrows that denote the causal influences among themselves. Each causal link has a polarity, either positive – denoted with the “+” sign – or negative – denoted with the “-” sign –, which indicates how the dependent variable (at the arrowhead) changes when the independent variable (at the tail of the arrow) changes. Each meaningful loop is marked by a loop identifier, which indicates the type of feedback: a reinforcing – identifier marked with the R letter – or balancing – identifier marked with the B letter. The red elements indicate exogenous determinants. Fig. 1 reports an example of causal-loop diagram, which describes the interrelation between the use of electricity and the operation of the local market.

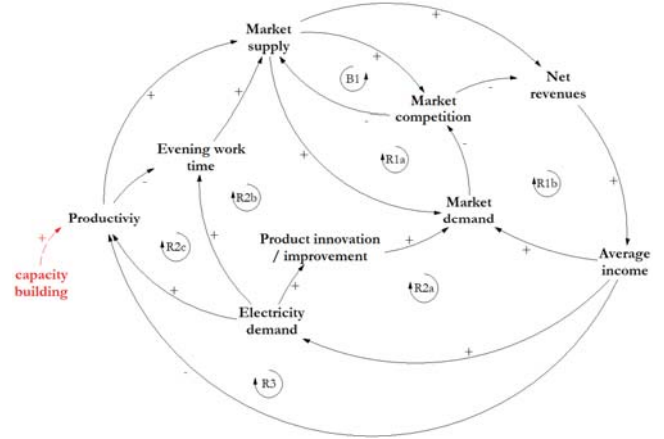


Figure 1. Causal-loop diagram regarding the use of electricity and the operation of the local market

As an explanation of the causal relations displayed in the diagram, are here reported few examples of reinforcing and balancing loops:

- *R2a: Electricity demand → Product innovation/improvement → Market demand → Market competition → Net revenues → Average income → Electricity demand.* The use of electricity causes an increase of market demand because of product innovation and quality improvement, increasing the sales, the revenues, and the income that can be reinvested in higher electricity consumption at work or at home.
- *R2b: Electricity demand → Evening work time → Market supply → Net revenues → Average income → Electricity demand.* The use of electricity for evening lighting allowed local IGAs to work more and increase their sales, the revenues, and the income that can be reinvested in more electricity consumption at work or at home.
- *R2c: Electricity demand → Productivity → Market supply → Net revenues → Average income → Electricity demand.* The electrification of local production because of electricity use (e.g. electric milling machines, carpentries, garages) increases the productivity, the sales, the revenues, and the income that can be reinvested in higher electricity consumption at work or at home.
- *B1: Market supply → Market competition → Market supply.* This is the fundamental balancing loop, which controls the *market equilibrium* between supply and demand.

In all the reinforcing loops R2, capacity building emerged to be an important factor for improving local productivity.

B. Formulation of the simulation model

According to Sterman [13], the formulation of a simulation model considers the specification of its mathematical structure – *i.e.* the conversion of the feedback diagrams to algebraic, differential, and integral equations –,

as well as the estimation of the parameters' value. The simulation framework is based on Vensim DSS ® software. For each discrete time-step dt , the software solves all the equations that describe the dynamic behaviour of the modelled system. Thus, the time bounds, and the integration method represent fundamental settings (TABLE I).

TABLE I. TIME BOUNDS AND INTEGRATION METHOD SET IN VENSIM ®

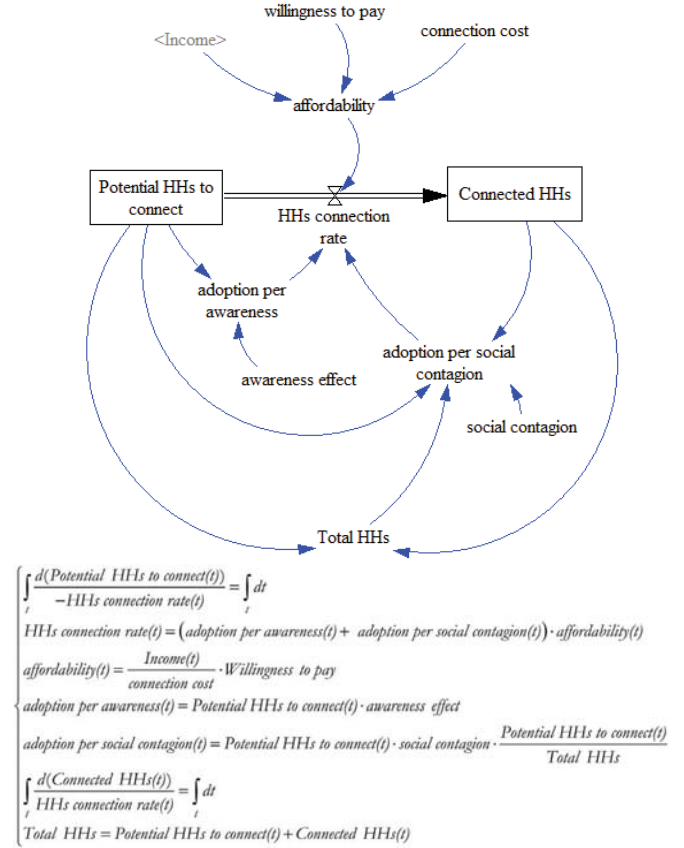
Time unit (t_u)	<i>Week</i>
Time step (Δt)	<i>0.25</i>
Initial time (Week)	<i>1 (January 2005)</i>
Final time (Week)	<i>679.25 (December 2017)</i>
Integration method	Euler: $Y_{t+dt} = \int_t^{t+dt} x \cdot dt = Y_t + x_t \cdot \Delta t$

The structure of the model is based on the causal-loop diagrams derived from the conceptualisation phase. Its formulation followed an iterative process based on the questionnaires filled in during the on-field experience, the information shared with the local experts, and the analysis and re-redefinition of the structure by the modeller. The final structure of the model is composed by 80 integral equations, 260 auxiliary equations (*i.e.* algebraic and differential), and 140 constants. The structure of the model is based on a stock-and-flow formulation. *Stocks* are integral accumulation variables, while *flows* are differential equations. All the other relations are based on algebraic auxiliary equations. The model simulates the impact of electricity access and its use on the socio-economic development experienced in Ikondo, and the related feedbacks on the community's electricity consumption. One of the outputs is therefore the projection of long-term electricity load curves of the entire community along the simulated time horizon.

An example of stock and flow structure and related mathematical formulation is represented in Fig. 2, which represents the dynamics of the diffusion of households' (HHs) connections. The phenomenon is based on the effect of advertising, social contagion, and affordability. It mathematically explains part of the qualitative relation *Average Income* $\rightarrow$ *Electricity demand* of Fig. 1. The formulation is an adaptation of the well-known Bass diffusion model [14], which models the spread of innovations in a social system through the representation of two main dynamics:

- The time-invariant tendency to adopt through external influences (*awareness effect*) – *e.g.* advertising and other communications independent from the social network.
- The social interaction (*social contagion*) that results from interactions among the agents of the social system – *viz.* adopters and potential adopters of an innovation.

The “affordability” aspect has been added to consider the effect of income on the diffusion of new electrical connections. The time-independent parameters are all calibrating parameters (see the following sub-section).



Another example reported below represents the dynamics that stand at the basis of goods and services supply in the local market. These are driven by the increase of productivity given by electricity use of each single IGA, and the time of operation. It mathematically explains part of the qualitative relations *Evening work time* $\rightarrow$ *Market supply*, *Electricity demand* $\rightarrow$ *Productivity* $\rightarrow$ *Market supply*, and *Capacity building* $\rightarrow$ *Productivity* of Fig. 1. In the field of economics, the representation of the production of goods and services relies on the so-called “production function”, which relates the physical output Q of a production process to quantities of inputs $X_1, X_2, \dots, X_n$. The Cobb-Douglas production function [15] is widely used to represent the technological relationship between two or more inputs and the total output that can be produced by those inputs. In its generalised form, the Cobb-Douglas function is proportional to the product of n terms representing the inputs of the production, each one to the power of its own output elasticity (1).

$$Q = A \prod_{i=1}^n X_i^{\lambda_i} \quad (1)$$

where λ_i is the output elasticity of input X_i , which measures the responsiveness of the output Q to a change in X_i . A is the total factor productivity. In this study, this formulation is used to express the market production of local businesses as the product between the working time and the productivity of each activity. The latter is expressed in terms of units of goods produced by each activity in the unit of time (hours), with the level of income as proxy of the physical capital. The total factor productivity A_{market} is a function of the electrification rate of the business activities and the effect of capacity building activities, as stated by CEFA's experts. According to local surveys, changes in productivity are not immediately generated by variations of production inputs; instead they occur after a time lag due to people's hesitation to new investments and the physical time needed to adapt their business to increase in the available capital and new knowledge. Fig. 3 shows the graphical representation and the mathematical formulation of the market production dynamics.

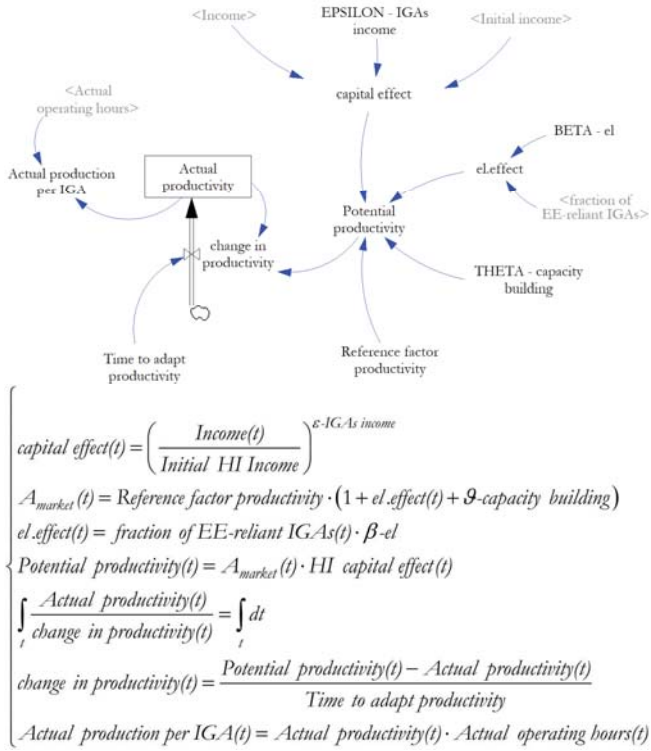


Figure 3. 2nd example of stock-and-flow diagram and the mathematical formulation

The *fraction of EE-reliant IGAs* variable represents the percentage of IGAs that are highly dependent on electricity use. Still, the time-independent parameters are all calibrating parameters (see the following sub-section).

C. Model Calibration

The model calibration is used here to assess the model structure by verifying its ability to replicate the observed historical behaviour of the system. For implementing the calibration, we relied on:

- 4 time-series (*i.e.* the number of metered electricity connections of households and businesses, and the monthly electricity consumptions of households and businesses) to set the payoff between the 4 data series and 4 corresponding output variables v of the model:

$$\text{payoff}_k = - \sum_{t=1}^T \sum_{v=1}^4 \left[\frac{(\text{model value}_v(t) - \text{datum}_v(t))^2}{\text{weight}_v} - a_v \right] \quad (2)$$

with the term a_v the natural logarithm of a function of weight_v , which is a calibrating parameter itself.

- local interviews, data, and literature to define the search space for each one of the 123 calibrating parameters;
- the Powell algorithm [16] to find the optimal solution.

D. Sof-linking demand and stochastic load models

For the calibrated projection, a long-term load scenario were generated by developing a soft-link with the stochastic load profile generator *LoadProGen* [11], [17], [18]. The software was modified to simulate several daily profiles in line with the duration of the desired scenario, and to automatically aggregate them. The aggregation was formulated in the original code by adding an external routine, which runs *LoadProGen* as many times as the number of the N days (*i.e.* the lifetime) of the desired scenario and according to a *calendar-file*, which associates each day of the scenario to a specific input data file. One input data file can represent one single day – in this way the number of input data files is equal to the number of days of the scenario – or a “cluster” of days (e.g. a season) – in this way the number of input files is equal to the number of such “clusters” of the scenario. This version is called *Lifetime-LoadProGen*.

IV. RESULTS AND DISCUSSION

A. Calibrated Results

For the sake of brevity, only the results concerning the historical fit on the variable representing the total electricity demand of the village are presented (Fig. 4). The goodness of the fit is assessed by relying on appropriate descriptive statistical measures reported in TABLE II (e.g. Mean Absolute Percent Error, Theil's Inequality, which is a decomposition of the Mean Square Error).

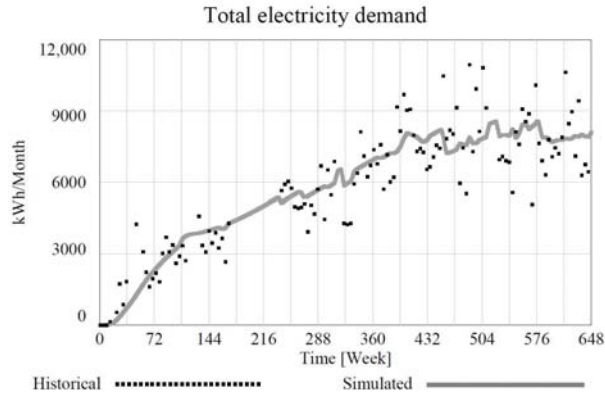


Figure 4. Historical fit between model output and data of the *Total electricity demand* variable

TABLE II. STATISTICS FOR THE GOODNESS OF FIT

MAPE=17%	The variability of data around the mean of the model output is almost 20%. This means that the model is not able to capture the point-by-point variability of data. This issue can be solved by (i) introducing and calibrating a random error term, or (2) by formulating the dynamics that can explain such variability. Potential reasons of this variability can be: – Intra-season variabilities, due to the dependency of people financial availability with the harvesting period; – Daily load variability, due to the stochastic nature of people's habits and use of electricity within the 24 hours. This high level of variability is not a systematic problem, since the purpose is not to study such short-term variabilities in the long-term forecast, and because intra-days uncertainties are specifically modelled through a stochastic load profile generator.
$U^M = 0\%$	<i>NO bias</i> : The model output and the data series have the same mean.
$U^S = 3\%$	<i>NO unequal variation</i> : The model output and the data series have quite the same trend.
$U^C = 97\%$	<i>ONLY unequal covariation</i> : The error term is exclusively attributable to an error term with zero mean, which makes the model output and the data series to differ point by point. It is an <i>unsystematic</i> error since the purpose is not to study cycles in the data.

The calibration allowed also to derive interesting insights on the dynamics affecting the total consumption of electricity:

- First, initial months show the trend typical of a S-shape diffusion curve, but with a short early-stage of adoption. This suggests that electricity access is not perceived as an actual innovation by people, but something that they already know and desire.
- Second, the change in the trend after the first 20/25 months caused by the change of the tariff scheme of the electrical connections has a significant impact on the electricity consumption. This indicates the importance of connection cost on the diffusion of the electrical connections.

B. Generation of stochastic long-term load profiles

Fig.5 reports the results for 2 random days at year 1 and 13 of the planning horizon.

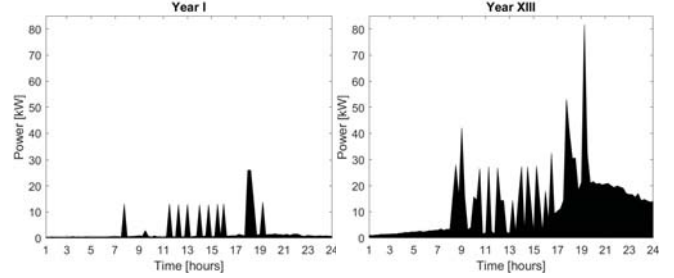


Figure 5. Load profiles for 2 random days at year 1 and 13 of the planning horizon

From the profiles, the following considerations can be derived:

- The consumption of businesses, especially the most energy intensive ones, clearly determines the power peak, suggesting the importance of assessing the evolution of business activities and their consumptions.
- The household consumption determines the area at the base level of the demand, as visible in the morning and evening hours. It increases with the years, due to the growth in the residential connections.
- Along with time load shapes come neater: a first peak in the morning, due to both starting household and IGAs activities; a little decrease during the day, since farmers are mainly busy in the land, and the only load is consumed by the IGAs; in the evening, due to the recovery of household activities at home, and the profile of the IGAs working in the evening thanks to the electrical lighting.

The simulated long-term load profile was tested *a-posteriori* to calculate the total electricity monthly demand, and then compared with the calibrated curve obtained with SD (Fig. 6). The *Lifetime-LoadProGen* curve is subjected to more variability because of the stochasticity algorithm of the software, which generates different daily curves at each run, and due to the randomness adopted to set the different user classes generated through the soft-link with the SD model.

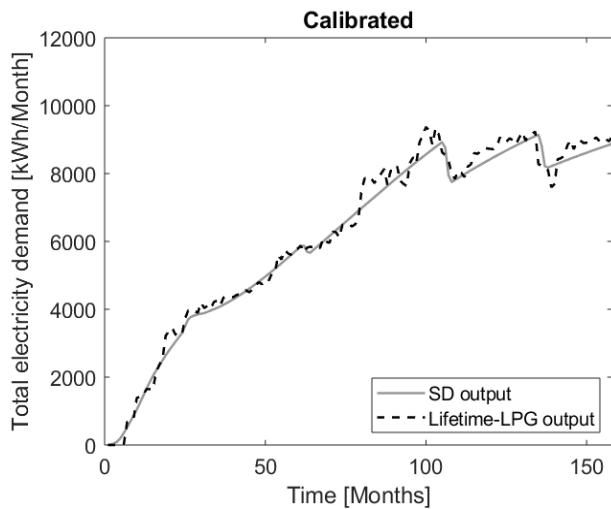


Figure 6. Long-term load profiles for the selected scenarios, and comparison with the SD output.

V. CONCLUSIONS

Long-term projections are an essential input for sizing off-grid power systems in rural areas of developing countries. On the other hand, the uncertainty of long-term forecasts depends on the complex interactions and dynamics involving the electricity use and socio-economic development. To introduce a more reliable method for generating long-term projections of electricity demand in rural areas, this study concerns the conceptualisation and the formulation of a System Dynamics (SD) calibrated with data gathered from a hydroelectric-based electrification programme implemented the rural community of Ikondo, Tanzania, in 2005. The model has been soft-linked with the stochastic load profile generator LoadProGen, to convert “demand” into “load” projections, which can be directly used as input in off-grid optimization software.

The formulation of the model and its testing allowed to identify and confirm some of the principal dynamics affecting the evolution of electricity demand (e.g. the testing process reveals that the electricity connections’ diffusion is mostly influenced by the cost rather than awareness of possible advantages of electricity use), which can support the definition of appropriate electricity-access oriented policies. Moreover, the calibration of the model confirms a good fitting with the historical available data, confirming the ability of the model to represent the main dynamics affecting the evolution of electricity demand in rural contexts. Further case studies are needed to develop a more “generalised” and “applicable” model. Nevertheless, in the relatively new exploratory research field on energy demand models for rural electrification, we believe that this work sets an important starting point and a reference modelling framework.

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